

FULL-STACK PROTOTYPE · EXACT LOOKUP · LANGGRAPH

FU Berlin CS Consultant

A study consultant for the **FU Berlin Master Informatik** programme. It answers programme questions from local resources and validates proposed study plans against the deterministic 2014 Master Informatik rules.

Next.js · React · TypeScript

FastAPI · Python

LangGraph

Exact Course Lookup

Docker Compose

1 What it does

A backend-first study consultant with a standalone Next.js frontend. The LLM parses and explains; deterministic Python rules decide whether a study plan is valid.

Programme Q&A

Answers rule questions from the structured rule catalogue and course-offering questions from exact JSON buckets.

Study-Plan Check

Parses a proposed plan, validates LP totals and area rules, and explains issues.

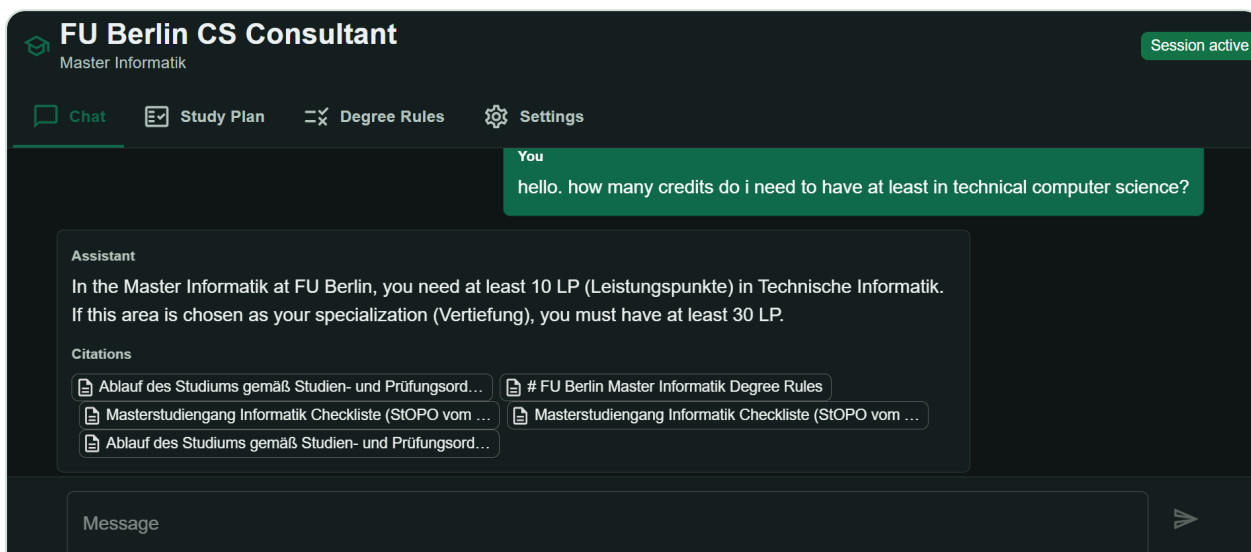
Degree Rules

Structured, read-only catalogue of programme rules served via a dedicated API.

Advisory only — official FU Berlin documents and the examination office remain authoritative for real study decisions.

2 Chat in action

A real session in the consultant frontend — question, grounded answer, citations, and the four-tab shell around it.



Consultant chat tab - grounded answer with citations from local course data.

3 Architecture

Two services orchestrated by Docker Compose.

SERVICE	STACK	ENDPOINT
Frontend	Next.js App Router · React · TypeScript · Material UI	<code>http://localhost:3000/consultant</code>
Backend	FastAPI · LangGraph · Pydantic	<code>http://localhost:5100</code>

Browser + Frontend

Next.js UI at `:3000` . The browser calls the backend through the host-facing API URL.

- Chat
- Study Plan
- Degree Rules
- Settings



FastAPI Backend

Owns sessions, transcript upload, LangGraph routing, rule validation, and response composition.

- LangGraph
- PDF parser
- Rule checker
- API



Local Sources

Course buckets, program-rule catalogue, deterministic degree rules, and in-memory session state.

LLM Provider

AcademicCloud, FU Ollama over SSH, or local Ollama depending on `LLM_PROVIDER` .

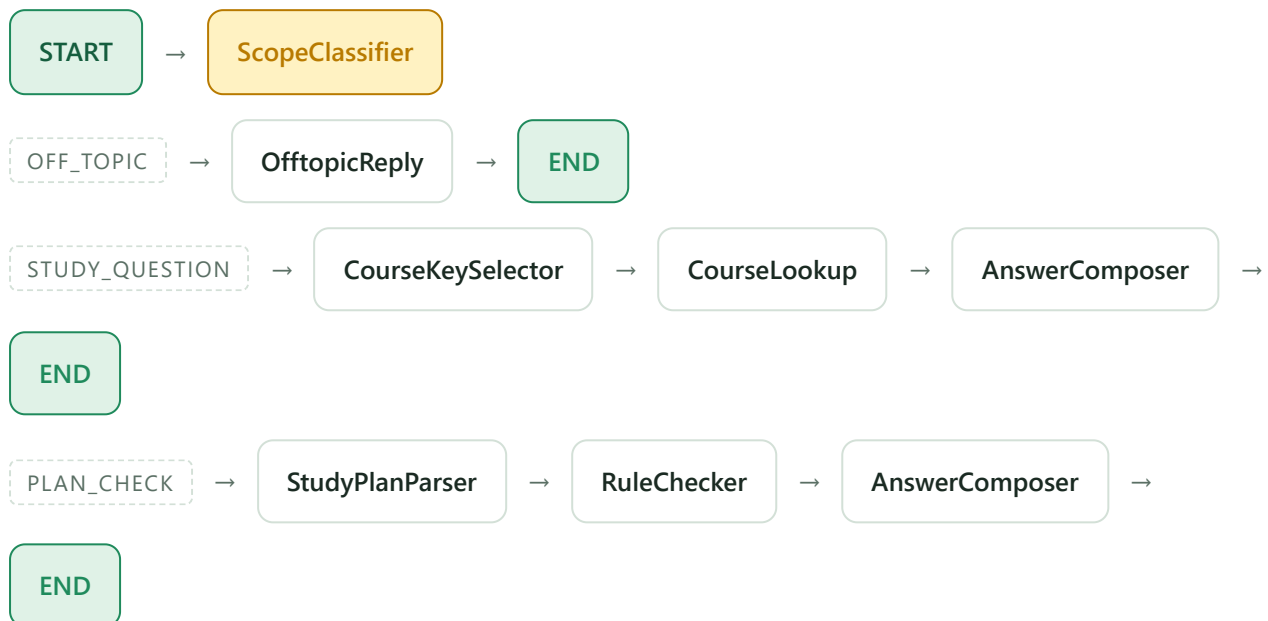
Important Docker detail: `NEXT_PUBLIC_API_BASE_URL` points to `http://localhost:5100` because frontend JavaScript runs in the user's browser, not inside the Docker network.

Backend API

```
GET /health
GET /api/program-rules
POST /api/sessions
POST /api/sessions/{session_id}/message
POST /api/sessions/{session_id}/transcript
```

4 Agent Flow

A LangGraph state machine routes every user message to the right pipeline. Off-topic messages short-circuit. Study questions go through exact course lookup when needed; plan checks skip retrieval and go through deterministic validation.



LLM parses → Python validates → LLM explains

CourseKeySelector may return zero, one, or many lookup buckets such as `sose26/technical/swp`. CourseLookup then returns the complete selected buckets from local JSON.

5 Study-Plan Validation

The rule checker is plain Python — auditable, deterministic, and never delegated to the LLM. It currently covers:

90 LP module area before thesis

30 LP Masterarbeit

70–80 LP Informatik area

10–20 LP Anwendungsbereich

Specialization minimums

Wissenschaftliches Arbeiten min/max

Softwareprojekt min/max

Wahlbereich seminar/project caveat

25–30 LP ungraded modules

Max 15 LP Bachelor modules

Duplicate module detection

6 Study Plan in action

The Study Plan tab renders the latest extracted plan with LP totals, the chosen Vertiefung, Wahlbereich data, and the per-area module breakdown.

The screenshot shows the 'Study Plan' tab in the 'FU Berlin CS Consultant' application. The header includes the application name, 'Master Informatik', and a 'Session active' status. Navigation links for 'Chat', 'Study Plan' (active), 'Degree Rules', and 'Settings' are present. The main content area displays the 'Study Plan' with a subtitle 'Latest extracted plan from your chat messages.' and a 'Save as PDF' button. Below this, a summary bar shows: 'Total: 120 LP', 'Vertiefung: Technische Informatik', 'Wiss. Arbeiten: 2', 'Softwareprojekt: 1', 'Unbenotet: 30 LP', and 'Bachelor: 0 LP'. The plan is organized into three columns: 'Praktische Informatik' (20 LP), 'Theoretische Informatik' (25 LP), and 'Technische Informatik' (35 LP). Each column lists modules with their respective LP values and status (e.g., 'Softwareprojekt' is 'unbenotet').

Area	Module	LP	Status
Praktische Informatik (20 LP)	Softwareprojekt Campus Planner B	10 LP	unbenotet
	Softwareprojekt	10 LP	unbenotet
	Advanced Software Engineering	10 LP	
Theoretische Informatik (25 LP)	Advanced Algorithms	10 LP	
	Model Checking	10 LP	
	Wissenschaftliches Arbeiten	5 LP	
	Theoretische Informatik	5 LP	unbenotet
Technische Informatik (35 LP)	Secure Systems	10 LP	
	Computer Architecture Lab	10 LP	unbenotet
	Machine Learning	10 LP	
	Secure Systems	10 LP	

Study Plan tab — extracted modules grouped by area with LP totals and Wahlbereich tags.

A **Save as PDF** button in the top-right exports the current study plan to a print-ready PDF for sharing with advisors or saving alongside official paperwork.

7 LLM Providers

Three interchangeable providers, selected via `LLM_PROVIDER`.

AcademicCloud

Hosted API, defaulting to `qwen3-235b-a22b`.

FU Ollama

FU `cuda01` reached automatically through an SSH tunnel.

Local Ollama

Ollama running on the host machine, reached via `host.docker.internal`.

8 Frontend

Next.js App Router with Material UI. The consultant app — not a marketing page — is the first screen. Four tabs cover the full workflow.

Chat

Session creation, message send, assistant response, citations and plan-check summaries.

Study Plan

Extracted modules, LP totals, Wahlbereich data and validation issues per student.

Degree Rules

Read-only rendering of `GET /api/program-rules`.

Settings

Dark mode plus developer tools — session ID, reset, backend health, API base URL.

9 Getting Started

Two commands to a running stack.

```
# 1. Configure the provider you want
cp .env.example .env

# 2. Build and start everything
docker compose up --build
```

Then open

Consultant — <http://localhost:3000/consultant>

API docs — <http://localhost:5100/docs>

Health — <http://localhost:5100/health>

10 Tech Stack

Next.js 14

React 18

TypeScript

Material UI

FastAPI

Pydantic

LangGraph

Ollama

Docker Compose

pytest

FU Berlin CS Consultant · prototype project · see [agent-flow.html](#) for the detailed routing diagram.